Regular KeyMap

Teensy I/O			8	7	19	4	6	5	22	23
	Keyboard		3	5	7	13	9	11	15	17
		RP2040 GP	6	5	7	2	4	3	8	9
32	30	24		*Ctrl-Left*					*Ctrl-Right*	
28	26	20			*Shift-Left*				*Shift-Right*	
26	22	18	Esc	^ TILDE	Tab	1	q	a	Ger:y / Intl:z KEY_Z	
31	18	23	Ger: <> | Intl:| NON_US_BS	F1	Caps-Locks	2	w	s	x	
17	14	15	F4	F2	F3	3	e	d	c	
25	10	17	g	5	t	4	r	f	v	b
33	6	26	F5	F9	Back-Space	F10		\ BACKSLASH	Enter	Space
15	2	13	h	6	Ger:z Intl:y KEY_Y	7	u	j	m	n
9	4	10	F6	´ EQUAL	}] RIGHT_BRACE	8	i	k	, COMMA	
16	8	14		F8	F7	9	o	l	. PERIOD	
27	12	19	Ger:ä Intl:" QUOTE	Ger:ß Intl:- MINUS	Ger:ü Intl:[{ LEFT_BRACE	0	p	Ger:ö Intl:; SEMICOLON	Ger:#' Intl:#~\ BACKSLASH	Ger:- Intl:/ SLASH
29	16	21	**ThinkVantage**	Delete	Vol-Down	F11	Vol-Up	Mute		Cursor Down
24	20	16		Insert		F12				Cursor Right
14	24	12		Page-Up		Page-Down	*GUI/Win*	Menu	Document-Left AC_BACK	Document-Right AC_FORWARD
30	28	22	Cursor Up	Home		End			Pause	Cursor Left
10	32	11	*Alt-Left*			Print-Screen	Scroll-Lock			*Alt-Right*

KeyTypes
Fn Key-Combo
NumPad
Added in ISO Missing in ANSI
ANSI-Key Missing in ISO
switched Ger/Eng
Multimedia
Modifier
Regular Key

Ignore the german swaps in green, every ISO-keyboard reports the same keycodes regardless of the keycaps. That is why you need to select the current keymap in your OS

Num-Lock

Teensy I/O			8	7	19	4	6	5	22	23
	Keyboard		3	5	7	13	9	11	15	17
		RP2040 GP	6	5	7	2	4	3	8	9
32	30	24		*Ctrl-Left*					*Ctrl-Right*	
28	26	20			*Shift-Left*				*Shift-Right*	
26	22	18	Esc	^ TILDE	Tab	1	q	a	Ger:y Intl:z KEY_Z	
31	18	23	Ger: <>| Intl:| NON_US_BS	F1	Caps-Lock	2	w	s	x	
17	14	15	F4	F2	F3	3	e	d	c	
25	10	17	g	5	t	4	r	f	v	b
33	6	26	F5	F9	Back-Space	F10		\ BACKSLASH	Enter	Space
15	2	13	h	6	Ger:z Intl:y KEY_Y	7 KEYPAD_7	4 KEYPAD_4	1 KEYPAD_1	0 KEYPAD_0	n
9	4	10	F6	´ EQUAL	}] RIGHT_BRACE	8 KEYPAD_8	5 KEYPAD_5	2 KEYPAD_2	, COMMA	
16	8	14		F8	F7	9 KEYPAD_9	6 KEYPAD_6	3 KEYPAD_3	. KEYPAD_PERIOD	
27	12	19	Ger:ä Intl:" QUOTE	Ger:ß Intl:- MINUS	Ger:ü Intl:[{ LEFT_BRACE	/ KEYPAD_SLASH	* KEYPAD_ASTERIX	- KEYPAD_MINUS	Ger:#' Intl:#~\ BACKSLASH	+ KEYPAD_PLUS
29	16	21	**ThinkVantage**	Delete	Vol-Down	F11	Vol-Up	Mute		Cursor Down
24	20	16		Insert		F12				Cursor Right
14	24	12		Page-Up		Page-Down	*GUI/Win*	Menu	Document-Left AC_BACK	Document-Right AC_FORWARD
30	28	22	Cursor Up	Home		End			Pause	Cursor Left
10	32	11	*Alt-Left*			Print-Screen	Scroll-Lock			*Alt-Right*

KeyTypes
Fn Key-Combo
NumPad
Added in ISO Missing in ANSI
ANSI-Key Missing in ISO
switched Ger/Eng
Multimedia
Modifier
Regular Key

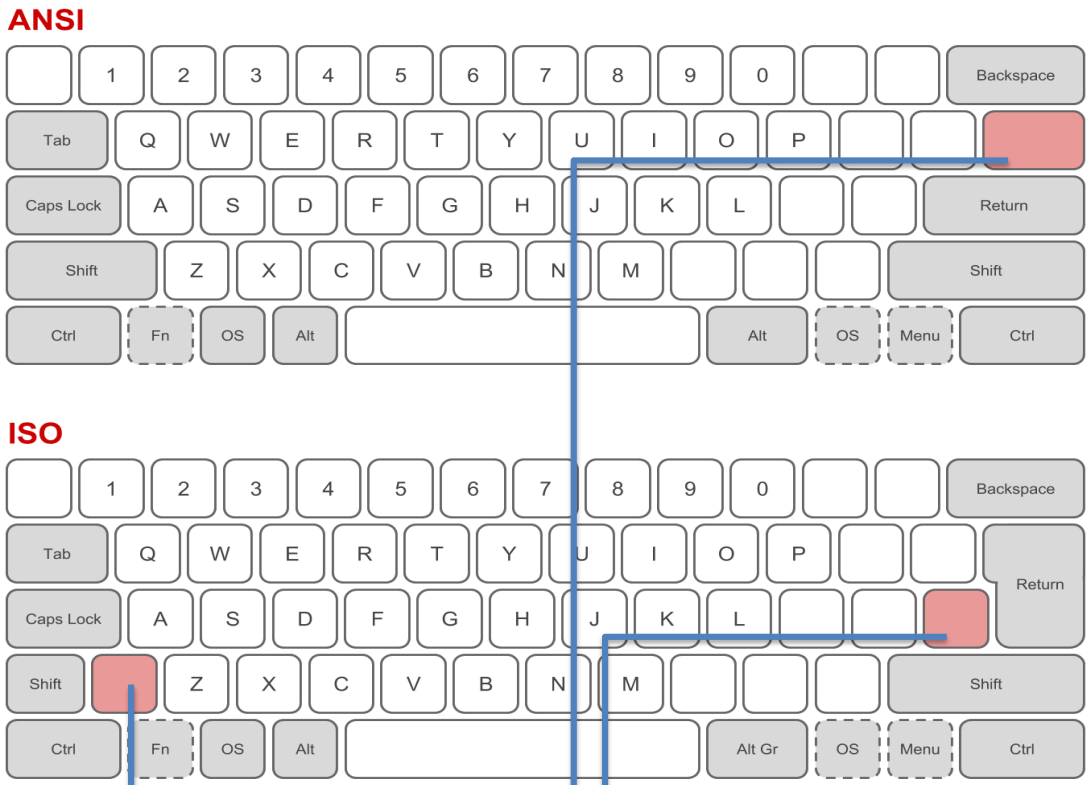
Ignore the german swaps in green, every ISO-keyboard reports the same keycodes regardless of the keycaps. That is why you need to select the current keymap in your OS

Fn-Overlay

Teensy I/O			8	7	19	4	6	5	22	23
	Keyboard P		3	5	7	13	9	11	15	17
		RP2040 GP	6	5	7	2	4	3	8	9
32	30	24								
28	26	20								
26	22	18								
31	18	23		F1						
17	14	15	F4 / Sleep	F2 / Lock *(Win+L)*	F3 / Screen Saver					
25	10	17								
33	6	26	F5 / Wifi-Toggle	F9 / Eject MEDIA_EJECT		F10				Space / Zoom
15	2	13				7 KEYPAD_7	4 KEYPAD_4	1 KEYPAD_1	0 KEYPAD_0	
9	4	10	F6			8 KEYPAD_8	5 KEYPAD_5	2 KEYPAD_2		
16	8	14		F8 Track Point/Pad	F7 / Project *(Win+P)*	9 KEYPAD_9	6 KEYPAD_6	3 KEYPAD_3	. KEYPAD_PERIOD	
27	12	19				/ KEYPAD_SLASH	* KEYPAD_ASTERIX	- KEYPAD_MINUS		+ KEYPAD_PLUS
29	16	21	**ThinkVantage**	Delete		F11				Play/Pause **MEDIA_PLAY_PAUSE**
24	20	16		Insert		F12 / Hibernate				Next Track **MEDIA_NEXT_TRACK**
14	24	12		ThinkLight					|< **MEDIA_REWIND**	>| **MEDIA_FAST_FORWARD**
30	28	22	Stop **MEDIA_STOP**	Screen Bright-ness Up		Screen Bright-ness Down			Pause	Prev Track **MEDIA_PREV_TRACK**
10	32	11				SysReq	Num-Lock			

KeyTypes
Fn Key-Combo
NumPad
Added in ISO Missing in ANSI
ANSI-Key Missing in ISO
switched Ger/Eng
Multimedia
Modifier
Regular Key

Ignore the german swaps in green, every ISO-keyboard reports the same keycodes regardless of the keycaps. That is why you need to select the current keymap in your OS



Notes	Keycodes	Ger+AltGr	Intl.+Shift	International	Ger+Shift	German	Intl.+AltGr
Column on Pin 3 Row in Pin 18	NON_US_BS **100 | 0xF000**	|	|	\	>	<	
Column on Pin 5 Row on Pin 6	BACKSLASH **49 | 0xF000**			\			
Column on Pin 15 Row on Pin 12	BACKSLASH **49 | 0xF000**		~	#	'	#	\

ANSI-Key Missing in ISO
Added in ISO Missing in ANSI

Identifier	KeyCode	Keys pressed	
AC_FORWARD	0x225 | 0xE400	Document right	works as Forward/Back in most applications (e.g. browser)
AC_BACK	0x224 | 0xE400	Document left	
AC_ZOOM	0x22F | 0xE400	Fn+Space	does not do anythink in Windows
BRIGHTNESS_INC	0x6F | 0xE400	Fn+Home	working on Windows
BRIGHTNESS_DEC	0x70 | 0xE400	Fn+End	but your screen needs to support changing the brightness over HDMI/DP
AL_CALCULATOR	0x192 | 0xE400	ThinkVantage	Calculator on ThinkVantage-Button, but could be anything else
MEDIA_MUTE	0xE2 | 0xE400	Mute	all working
MEDIA_VOLUME_DEC	0xEA | 0xE400	Volume Down	
MEDIA_VOLUME_INC	0xE9 | 0xE400	Volume Up	
MEDIA_PREV_TRACK	0xB6 | 0xE400	Fn+Left	all working
MEDIA_NEXT_TRACK	0xB5 | 0xE400	Fn+Right	
MEDIA_PLAY_PAUSE	0xCD | 0xE400	Fn+Down	
MEDIA_STOP	0xB7 | 0xE400	Fn+Up	
MEDIA_FAST_FORWARD	0xB3 | 0xE400	Fn+DocRight	working on windows
MEDIA_REWIND	0xB4 | 0xE400	Fn+DocLEft	using these two keys additionally to the blue imprints for media control
AL_TERMINAL_LOCK	0x19E | 0xE400	Fn+F2	does not work on Windows, send Win+L instead
AL_SCREEN_SAVER	0x1B1 | 0xE400	Fn+F3	does not do anythink in Windows
SYSTEM_SLEEP	0x82 | 0xE200	Fn+F4	works on Windows
AL_WIFI_STATUS	0x1AD | 0xE400	Fn+F5	does not do anythink in Windows
SYSTEM_PROJECT	0xB5 | 0xE200	Fn+F7	does not work on Windows, send Win+P instead
		Fn+F8	used internally to switch TrackPoint on/off
MEDIA_EJECT	0xB8 | 0xE400	Fn+F9	does not seem to work
SYSTEM_HIBERNATE	0x81 | 0xE200	Fn+F12	works on Windows

```
// Keycodes-Ranges:
//     32 - 127     ASCII direct (U+0020 to U+007F) <-- uses layout
//    128 - 0xC1FF  Unicode direct (U+0080 to U+C1FF) <-- uses layout
// 0xC200 - 0xDFFF  Unicode UTF8 packed (U+0080 to U+07FF) <-- uses layout
// 0xE000 - 0xE0FF  Modifier key (bitmap, 8 keys, shift/ctrl/alt/gui)
// 0xE200 - 0xE2FF  System key (HID usage code, within usage page 1)
// 0xE400 - 0xE7FF  Media/Consumer key (HID usage code, within usage page 12)
// 0xF000 - 0xFFFF  Normal key (HID usage code, within usage page 7)
```